Quine - McCluskey Method - (or Tabular Method)

The Quine - McCluskey method, also known as the tabular method, is a more systematic method of minimizing expressions of even larger number of variables.

This is suitable for hand computation as well as computation by machines, i.e., it is programmable.

The fundamental idea on which this tabulation procedure is based is that, repeated application of the combining theorem $PA + P\bar{A} = P$ on all adjacent pairs of terms, yields the set of all prime implicants, from which a minimal sum may be selected.

each has 3 literals per term

ex:- $\Sigma m(0,1,4,5) = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C$

$= \bar{A}\bar{B}(C+\bar{C}) + A\bar{B}(C+\bar{C})$

Two terms of 2 literals per term are combined

$= \bar{B}(A+\bar{A})$

$= \bar{B}$

Steps for minimization of Boolean expression -

step 1: List all the minterms expressed by its binary representation.

step 2: Arrange all minterms in groups of the same no. of 1s in column 1. Start with the least no. of 1s group & continue with groups of increasing no. of 1s.

The no. of 1's is called index / weight of the term.

step 3: Compare each term in a group of index i with every term in the group of index i+1 (succeeding group) & combine them using combining theorem.

Two terms from adjacent groups are combinable, if their binary representations differ by just a single digit in the same position; the combined terms consists of the original fixed representation with the differing one replaced by a dash (-).

Place a check mark (✓) next to every term, which has been combined with atleast one term (only single ✓ for a term which has been combined several times)

step 4 : Write the combined terms in column 2.
Repeat this by comparing each terms generated in step 3 ;
combine two terms which differ by only a single 1 and whose dashes are in the same position to generate a new term. Two terms with dashes in different positions cannot be combined.

Write the new terms in column 3 and put a check mark next to each term which has been combined in column 2.

Continue the process with terms in columns 3, 4 etc. until no further combinations are possible.

step 5 : The remaining unchecked terms constitute the set of prime implicants. (They are called PI b'coz they are not covered by any other term with fewer literals).

step 6 : List all the PIs and draw the prime implicant chart (don't cares if any should not appear in the prime implicant chart).

Don't cares

* During the process of generating PIs, don't care combinations are regarded as true combinations i.e. combinations for which the expression assumes a 1.
However, they are not considered in selecting a minimum set of PIs (not listed as columns in prime implicant chart).
By not listing them, we actually leave the specification of don't care terms open.

step 7 : Obtain essential prime implicants and write the minimal expression.

The prime Implicant Chart -

1. It is a pictorial representation of the relationships b/w PIs & the minterms of the expression.
2. It consists of an array of u rows & v columns where, u - no. of prime implicants
   v - " " minterms for which expression takes value 1.
3. ith row contains Xs placed at the intersections with the columns corresponding to minterms covered by ith prime implicant.
4. X (don't cares) are not entered in the chart.

Essential Prime Implicants -

1. These implicants will definitely occur in the final expression.
2. Any row in a PI chart which has atleast one minterm that is not present in any other row is called the essential row & the corresponding PI is called EPI.
3. Any column with a single X, PI corresponding to the row in which X appears is the EPI.

once EPI is selected, all minterms it covers are checked off.

If all the minterms are covered, union of all EPIs yields the minimal expression; otherwise additional PIs are necessary. find the minimal set of PIs from that.

Two rows (or columns) I & J of a prime implicant chart which have Xs in exactly the same columns (or rows) are said to be equal.

$I = J$

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Example: Obtain the minimal expression for $f = \Sigma m(1,2,3,5,6,7,8,9,12,13,15)$ using the tabular method.

① Obtain prime implicants —

Column 1

Index	Minterm	Binary
1	1	0001 ✓
	2	0010 ✓
	8	1000 ✓
2	3	0011 ✓
	5	0101 ✓
	6	0110 ✓
	9	1001 ✓
	12	1100 ✓
3	7	0111 ✓
	13	1101 ✓
4	15	1111 ✓

Column 2

Pairs	A B C D
(1, 3)	0 0 – 1 ✓
(1, 5)	0 – 0 1 ✓
(1, 9)	– 0 0 1 ✓
(2, 3)	0 0 1 – ✓
(2, 6)	0 – 1 0 ✓
(8, 9)	1 0 0 – ✓
(8, 12)	1 – 0 0 ✓
(3, 7)	0 – 1 1 ✓
(5, 7)	0 1 – 1 ✓
(5, 13)	– 1 0 1 ✓
(6, 7)	0 1 1 – ✓
(9, 13)	1 – 0 1 ✓
(12, 13)	1 1 0 – ✓
(7, 15)	– 1 1 1 ✓
(13, 15)	1 1 – 1 ✓

Column 3

Quads		A B C D
(1, 3, 5, 7)	T	0 – – 1
(1, 5, 9, 13)	S	– – 0 1
(2, 3, 6, 7)	R	0 – 1 –
(8, 9, 12, 13)	Q	1 – 0 –
(5, 7, 13, 15)	P	– 1 – 1

Prime Implicants $= BD + A\bar{C} + \bar{A}C + \bar{C}D + \bar{A}D$

② Draw Prime Implicant chart —
↳ It consists of 11 columns corresponding to the minterms & 5 rows corresponding to the PIs – P, Q, R, S, T.

PIs \ Minterms	1	2 ✓	3 ✓	5 ✓	6 ✓	7 ✓	8 ✓	9 ✓	12 ✓	13 ✓	15 ✓
* P (5, 7, 13, 15)				X		X				X	X
* Q (8, 9, 12, 13)							X	X	X	X	
* R (2, 3, 6, 7)		X	X		X	X					
S (1, 5, 9, 13)	X			X				X		X	
T (1, 3, 5, 7)	X		X	X		X					

only P covers m15

only Q covers m12

EPI b'coz only R covers m2 & m6

From the above PI chart, we observe that

m_2 & m_6 — are covered by R only. So, R is an EPI.
Check off all minterms covered by R
(m_2, m_3, m_6 & m_7)

Similarly, m_8, & m_{12} — covered by Q only. So, Q is an EPI
Check off (m_8, m_9, m_{12} & m_{13})

Also, m_{15} — covered by P only. So, P is an EPI.
Check off (m_5, m_7, m_{13} & m_{15})

Now, only m_1 is not covered. Either S or T can cover it & both have same no. of literals.

∴, two minimal expressions are possible -

$$P+Q+R+S = BD + A\bar{C} + \bar{A}C + \bar{C}D$$

$$P+Q+R+T = BD + A\bar{C} + \bar{A}C + \bar{A}D$$

Ques. Using the tabular method, obtain the minimal expression for:-

$$f = \Sigma m(6,7,8,9) + d(10,11,12,13,14,15)$$

PIs = P → (8, 9, 10, 11, 12, 13, 14, 15) (1, 2, 4) dashes

8 4 2 1
1 - - -
A

& Q → (6, 7, 14, 15) (1, 8) dashes

\- 1 1 -
BC

Prime Implicant chart

PIs \ Minterms	6	7	8	9
*P → 8, 9, 10, 11, 12, 13, 14, 15			X	X
*Q → 6, 7, 14, 15	X	X		

both are EPIs

∴, minimal expression is

$$P + Q = \boxed{A + BC}$$